

JACKSON BOPP

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EDUCATION

University of South Florida

B.S. Computer Engineering, Dean's List, Gold's Scholars | GPA: 3.52

Tampa, FL

Fall 2026

Florida SouthWestern State College

High School Diploma, Associate in Arts, Dean's List | GPA: 3.86

Punta Gorda, FL

Fall 2022

PROJECTS

Testbench | Next.js, TypeScript, Python, PostgreSQL, MQTT, Docker, IBM watsonx.ai, Gemini

Spring 2026

- Built a full-stack hardware test automation platform streaming live telemetry from an MSP430 microcontroller via Raspberry Pi Zero 2 W over MQTT to a Next.js/TypeScript dashboard with real-time SSE
- Integrated IBM watsonx.ai (Granite-3-8b-instruct) for AI-powered failure analysis and Google Gemini 2.0 Flash as an embedded context-aware AI chat assistant with full REST API and Drizzle ORM
- Containerized full stack with Docker Compose (PostgreSQL + Mosquitto broker); wrote C firmware for MSP430 and a Python UART-to-MQTT bridge for the Raspberry Pi

Edward, AI Desktop Agent | Python, PyQt6, IBM Bob API, ElevenLabs, faster-whisper, FastAPI, PyAutoGUI, SQLite

Spring 2026

- Built the PyQt6 overlay UI (Iron Man palette, sliding panel, alchemy circle animation, listening indicator), ElevenLabs TTS/faster-whisper STT voice pipeline, smart clipboard and context enhancement system, and encrypted password vault wired into the system tray
- Implemented computer control system (mouse/keyboard automation via PyAutoGUI + FastAPI local API with confirmation handler), hybrid AI backends (moondream vision + llama text local pipeline and Gemini variant), Docker setup, and settings dialog

VoiceLegacy | Next.js 16, Tailwind CSS, Clerk, MongoDB Atlas, ElevenLabs, Gemini API, Vercel

Spring 2026

- Led all frontend development: landing page copy, CTAs, and animations; phrase bank layout, category filters, tone settings, and empty states; Speak For Me AI rewrite flow and action buttons; mobile layout polish across all pages
- Built local onboarding flow, record page controls, local Gemini API fallback, and managed PR review and merges throughout the hackathon

Raspberry Pi NAS | Raspberry Pi Zero 2 W, Samba, Linux, Python

Fall 2025

- Configured a Raspberry Pi Zero 2 W as a multi-user NAS using Samba for cross-platform file sharing with persistent Linux mount points, user authentication, and file-level permission management
- Implemented secure multi-user access controls and configured Linux services for reliable startup across reboots

OPEN SOURCE

- OpenBMC (phosphor-user-manager):** Root-caused a reported test failure to a missing test-environment dependency, then found and fixed two real bugs uncovered once the environment was corrected
- Zephyr RTOS:** Found and fixed a devicetree macro bug in the MIPI DBI display driver; added a permanent regression test so it can't silently reappear
- PlatformIO Core:** Reproduced a serial device monitor crash triggered by non-interactive input with a new regression test, then fixed it; merged upstream

EXPERIENCE

Smart City Student Volunteer

City of Winter Haven Technology Services

April 2026 - Present

Winter Haven, FL

- Assisted with physical server hardware installation and maintenance, IT support ticketing, and early-stage AI hosting infrastructure

Manager, Shift Leader & Front-End Web Support

Subs N' Such

April 2023 - March 2026

Lutz, FL

- Built and maintained a Square-based website for online ordering; managed shift operations and enforced food safety procedures

TECHNICAL SKILLS

AI / LLM Tools: Anthropic Claude API, Google Gemini API, IBM watsonx.ai (Granite), ElevenLabs, Ollama, OpenWakeWord

Languages: Python, TypeScript, JavaScript, C, C++, C#, MATLAB, R, Verilog/VHDL, RISC-V Assembly

Frameworks & Libraries: Next.js, React, FastAPI, PyQt6, Tailwind CSS, NumPy, SciPy, Matplotlib

Tools & Infrastructure: Docker, MQTT, PostgreSQL, MongoDB, SQLite, Git, Cursor, Raspberry Pi, Arduino, FPGA, Linux

Certifications: Certified SOLIDWORKS Associate (CSWA), CSWA Additive Manufacturing

INVOLVEMENT

Member, Society of Hispanic Professional Engineers (SHPE), Society of Asian Scientists and Engineers (SASE), and IEEE